

Vector algebra

2.2.5 GRADIENT OF A SCALAR FIELD

The *gradient* of p , ∇p , at a given point in space is defined as a vector such that:

1. Its magnitude is the maximum rate of change of p per unit length of the coordinate space at the given point.
2. Its direction is that of the maximum rate of change of p at the given point.

$$p = p(x, y, z)$$

$$\nabla p = \frac{\partial p}{\partial x} \mathbf{i} + \frac{\partial p}{\partial y} \mathbf{j} + \frac{\partial p}{\partial z} \mathbf{k}$$

$$p = p(r, \theta, z)$$

$$\nabla p = \frac{\partial p}{\partial r} \mathbf{e}_r + \frac{1}{r} \frac{\partial p}{\partial \theta} \mathbf{e}_\theta + \frac{\partial p}{\partial z} \mathbf{e}_z$$

$$p = p(r, \theta, \Phi)$$

$$\nabla p = \frac{\partial p}{\partial r} \mathbf{e}_r + \frac{1}{r} \frac{\partial p}{\partial \theta} \mathbf{e}_\theta + \frac{1}{r \sin \theta} \frac{\partial p}{\partial \Phi} \mathbf{e}_\Phi$$

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2.2.6 DIVERGENCE OF A VECTOR FIELD

Read the next

Cartesian: $\mathbf{V} = \mathbf{V}(x, y, z) = V_x \mathbf{i} + V_y \mathbf{j} + V_z \mathbf{k}$

$$\nabla \cdot \mathbf{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$$

Cylindrical: $\mathbf{V} = \mathbf{V}(r, \theta, z) = V_r \mathbf{e}_r + V_\theta \mathbf{e}_\theta + V_z \mathbf{e}_z$

$$\nabla \cdot \mathbf{V} = \frac{1}{r} \frac{\partial}{\partial r} (r V_r) + \frac{1}{r} \frac{\partial V_\theta}{\partial \theta} + \frac{\partial V_z}{\partial z}$$

Spherical: $\mathbf{V} = \mathbf{V}(r, \theta, \Phi) = V_r \mathbf{e}_r + V_\theta \mathbf{e}_\theta + V_\Phi \mathbf{e}_\Phi$

$$\nabla \cdot \mathbf{V} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 V_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (V_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial V_\Phi}{\partial \Phi}$$

A scalar quantity given as a function of coordinate space and time t is called a *scalar*
vector quantity given as a function of coordinate space and time is called a *vector*

Relations Between Line, Surface, and Volume Integrals

Consider again the open area S bounded by the closed curve C , as shown in Figure 2.11. Let $\mathbf{A}$ be a vector field. The line integral of $\mathbf{A}$ over C is related to the surface integral of $\mathbf{A}$ over S by *Stokes' theorem*:

$$\oint_C \mathbf{A} \cdot d\mathbf{s} = \iint_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S} \quad (2.25)$$

Consider again the volume $\mathcal{V}$ enclosed by the closed surface S , as shown in Figure 2.12. The surface and volume integrals of the vector field $\mathbf{A}$ are related through the *divergence theorem*:

$$\oiint_S \mathbf{A} \cdot d\mathbf{S} = \iiint_{\mathcal{V}} (\nabla \cdot \mathbf{A}) dV \quad (2.26)$$

If p represents a scalar field, a vector relationship analogous to Equation (2.26) is given by the *gradient theorem*:

$$\oiint_S p d\mathbf{S} = \iiint_{\mathcal{V}} \nabla p dV \quad (2.27)$$

MOMENTUM EQUATION

$$\frac{\partial}{\partial t} \iiint_V \rho \mathbf{V} dV + \iint_S (\rho \mathbf{V} \cdot \mathbf{dS}) \mathbf{V} = - \iint_S p \mathbf{dS} + \iiint_V \rho \mathbf{f} dV + \mathbf{F}_{\text{viscous}}$$

Momentum Eq.

$$- \iint_S p \mathbf{dS} = - \iiint_V \nabla p dV \quad \text{Gradient Theorem}$$

$$\iiint_V \frac{\partial(\rho \mathbf{V})}{\partial t} dV + \iint_S (\rho \mathbf{V} \cdot \mathbf{dS}) \mathbf{V} = - \iiint_V \nabla p dV + \iiint_V \rho \mathbf{f} dV + \mathbf{F}_{\text{viscous}}$$

[2.66]

the x component of Equation (2.66) is

$$\iiint_V \frac{\partial(\rho u)}{\partial t} dV + \iint_S (\rho \mathbf{V} \cdot \mathbf{dS}) u = - \iiint_V \frac{\partial p}{\partial x} dV + \iiint_V \rho f_x dV + (F_x)_{\text{viscous}}$$

[2.67]

$$\iint_S (\rho \mathbf{V} \cdot \mathbf{dS}) u = \iint_S (\rho u \mathbf{V}) \cdot \mathbf{dS} = \iiint_V \nabla \cdot (\rho u \mathbf{V}) dV$$

$$\iiint_V \left[\frac{\partial(\rho u)}{\partial t} + \nabla \cdot (\rho u \mathbf{V}) + \frac{\partial p}{\partial x} - \rho f_x - (\mathcal{F}_x)_{\text{viscous}} \right] dV = 0$$

MOMENTUM EQUATION

$$\frac{\partial(\rho u)}{\partial t} + \nabla \cdot (\rho u \mathbf{V}) = -\frac{\partial p}{\partial x} + \rho f_x + (\mathcal{F}_x)_{\text{viscous}}$$

$$\frac{\partial(\rho v)}{\partial t} + \nabla \cdot (\rho v \mathbf{V}) = -\frac{\partial p}{\partial y} + \rho f_y + (\mathcal{F}_y)_{\text{viscous}}$$

$$\frac{\partial(\rho w)}{\partial t} + \nabla \cdot (\rho w \mathbf{V}) = -\frac{\partial p}{\partial z} + \rho f_z + (\mathcal{F}_z)_{\text{viscous}}$$

Note that Equations (2.64) and (2.70a to c) apply to the unsteady, three-dimensional flow of any fluid, compressible or incompressible, viscous or inviscid. Specialized to a steady ($\partial/\partial t \equiv 0$), inviscid ($\mathbf{F}_{\text{viscous}} = 0$) flow with no body forces ($\mathbf{f} = 0$), these equations become

$$\oint\oint\oint_S (\rho \mathbf{V} \cdot d\mathbf{S}) \mathbf{V} = -\oint\oint\oint_S p d\mathbf{S} \quad \mathbf{[2.71]}$$

and

$$\nabla \cdot (\rho u \mathbf{V}) = -\frac{\partial p}{\partial x} \quad \mathbf{[2.72a]}$$

$$\nabla \cdot (\rho v \mathbf{V}) = -\frac{\partial p}{\partial y} \quad \mathbf{[2.72b]}$$

$$\nabla \cdot (\rho w \mathbf{V}) = -\frac{\partial p}{\partial z} \quad \mathbf{[2.72c]}$$